

ADAPTIVE TRAFFIC INTELLIGENCE FOR HIGH-PERFORMANCE LPX SYSTEMS



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PROBLEM STATEMENT

DYNAMIC WORKLOADS, UNEVEN LOADS,
LIMITED EFFICIENCY.



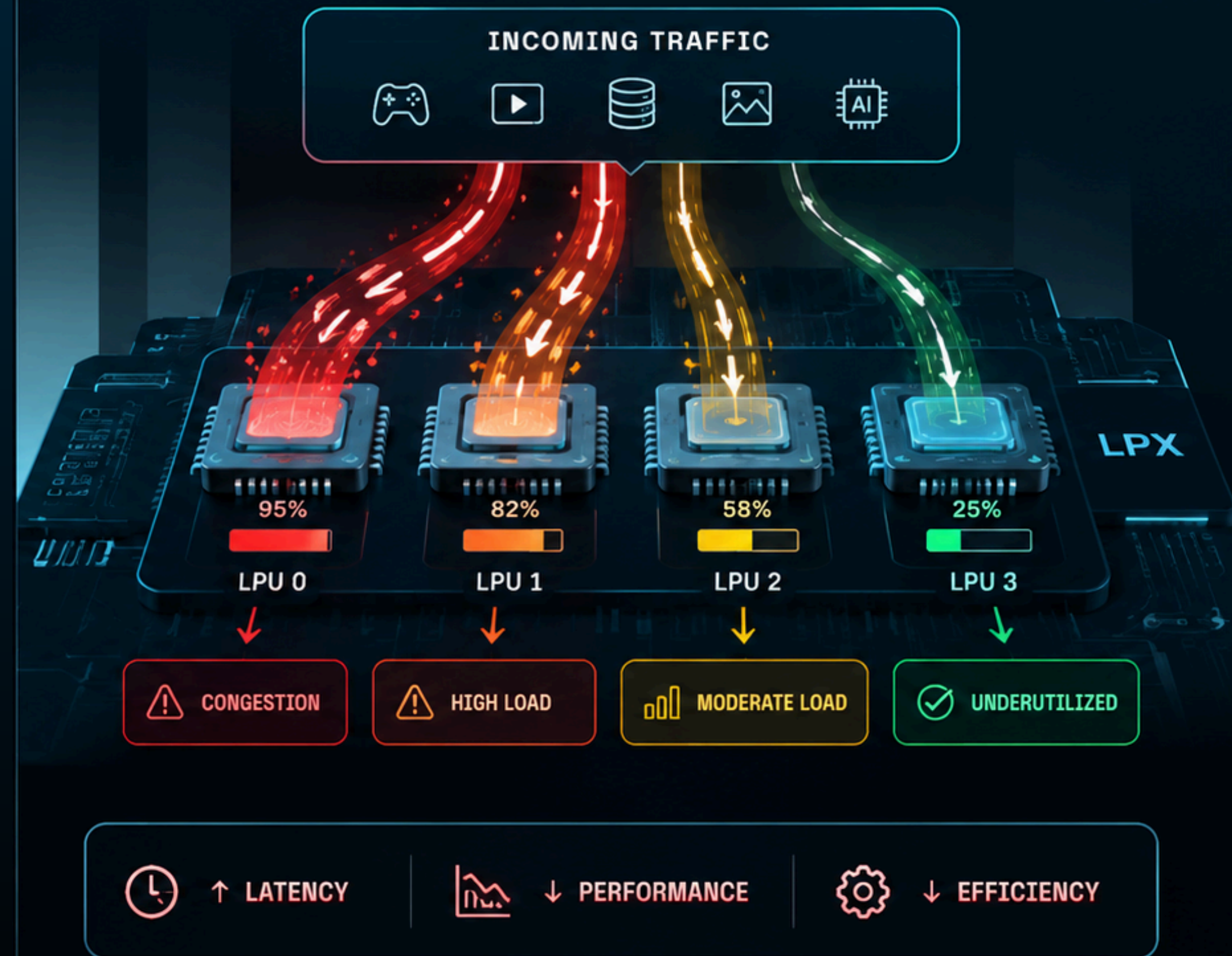
Modern high-performance LPX systems must handle **dynamic and unpredictable traffic workloads** across multiple processing units.



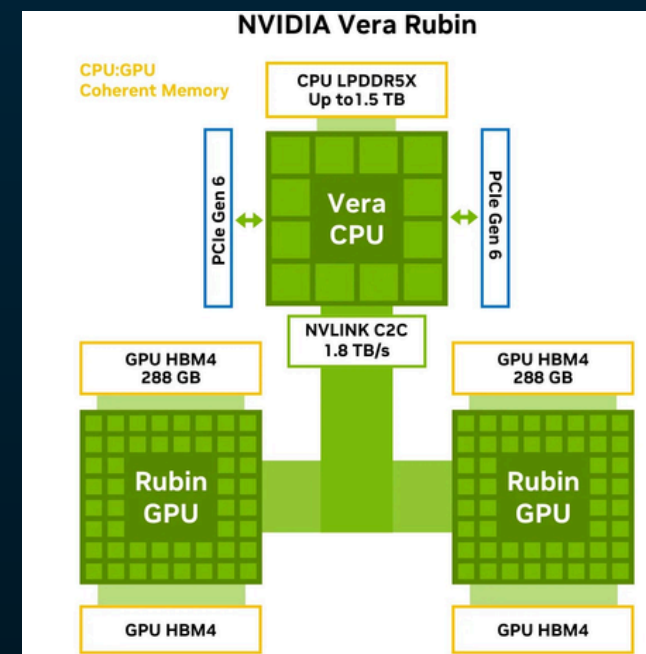
Conventional static task-mapping and routing mechanisms cannot effectively respond to sudden load variations leading to **congestion and uneven resource utilization**



This results in **increased latency, overload processing units**, and reduced overall system performance and efficiency



EXISTING LPX ARCHITECTURE

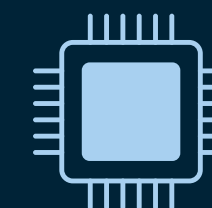
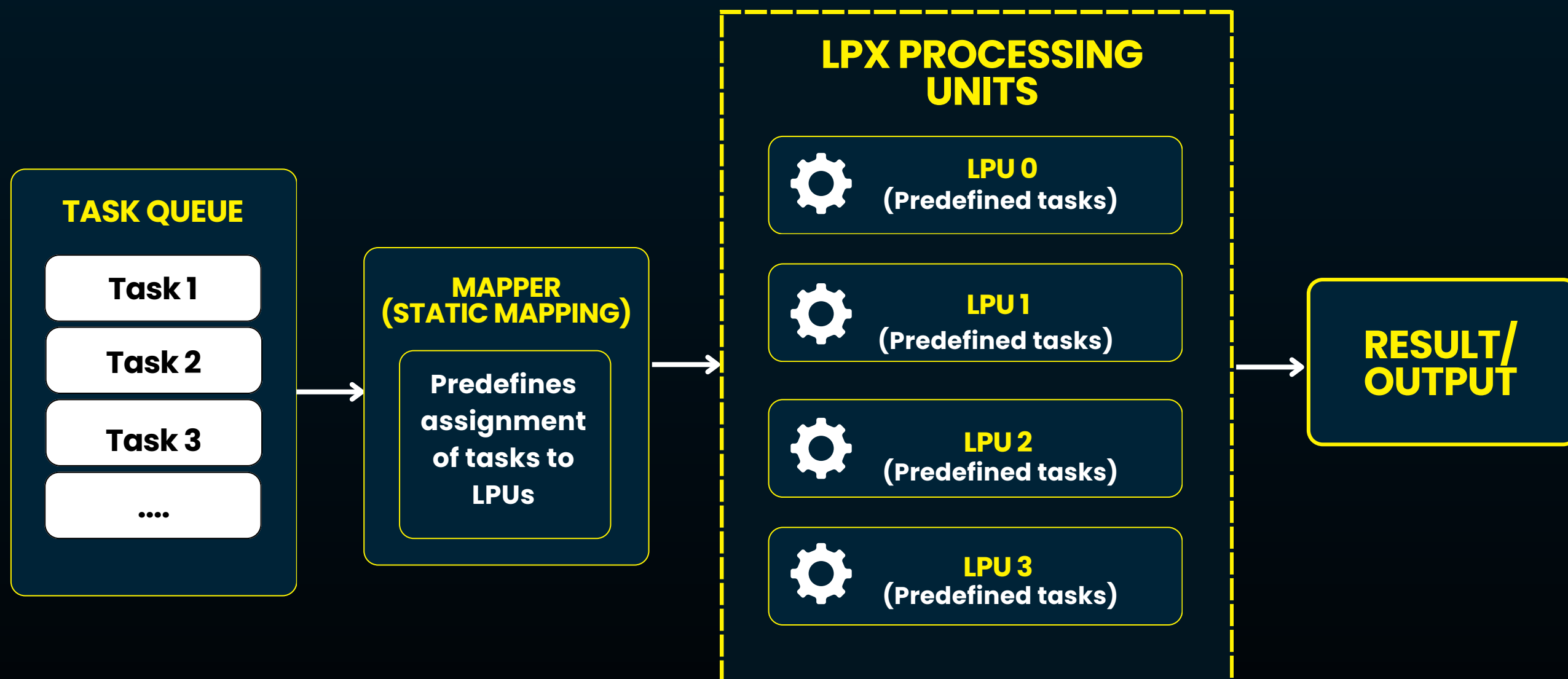


CURRENT LPX SYSTEM

LPX uses multiple processing units to execute incoming workloads in parallel, providing high computational throughput for demanding applications

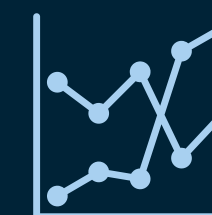
STATIC TRAFFIC & RESOURCE MAPPING

Current Architecture



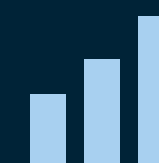
MULTIPLE LPUS

Incoming tasks are distributed across multiple Logical Processing Units for parallel execution



STATIC TASK MAPPING

Tasks follow predefined routing and allocation rules without considering real time workload conditions.



TRAFFIC BOTTLENECKS

Dynamic workloads can overload certain LPUs while others remain underutilized, causing congestion and increased latency

OUR PROPOSED SOLUTION

KEY IDEAS



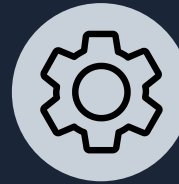
Monitor Load

Continuously observe LPU utilization and traffic patterns



Detect Stress

Identify congestion or overload conditions in real time



Adaptive Mapping

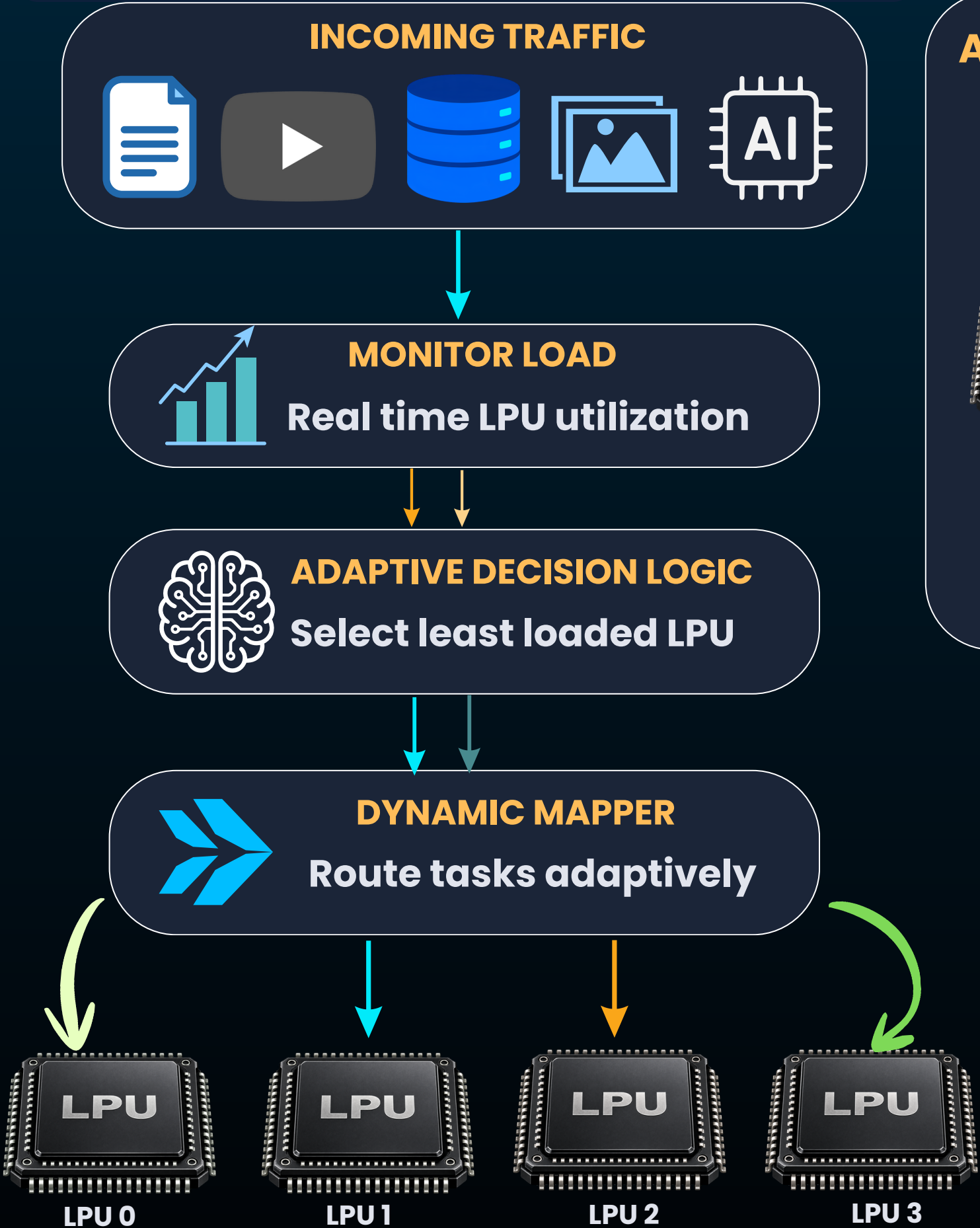
Dynamically redirect tasks to the least loaded LPU



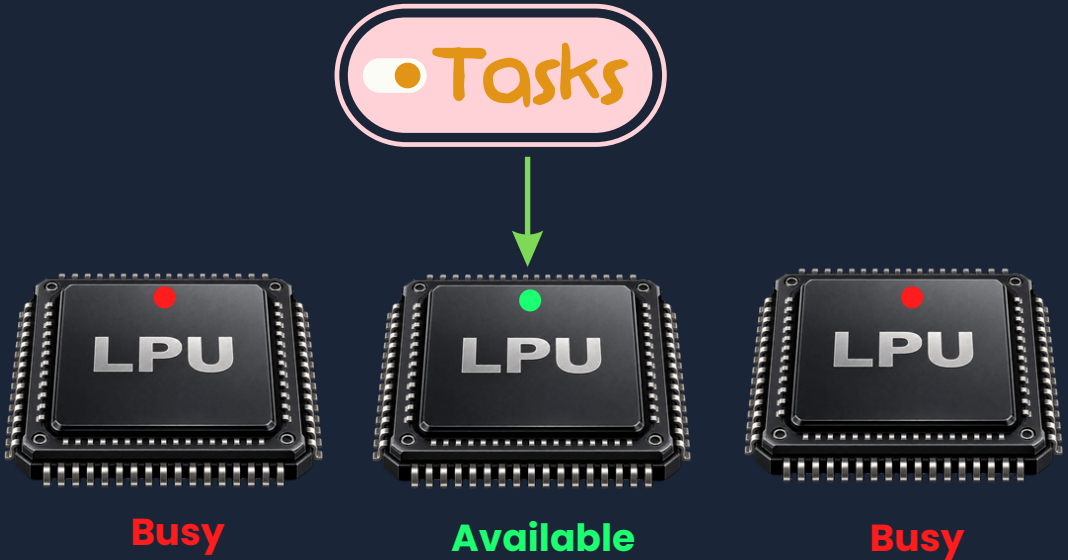
Improve Performance

Achieve balanced resource utilization and lower latency

Adaptive Stress-Aware Traffic Management



ADAPTIVE TRAFFIC MANAGEMENT



Checks each LPU's workload and sends tasks to an available unit

GOAL



Less Waiting



Balanced Workload

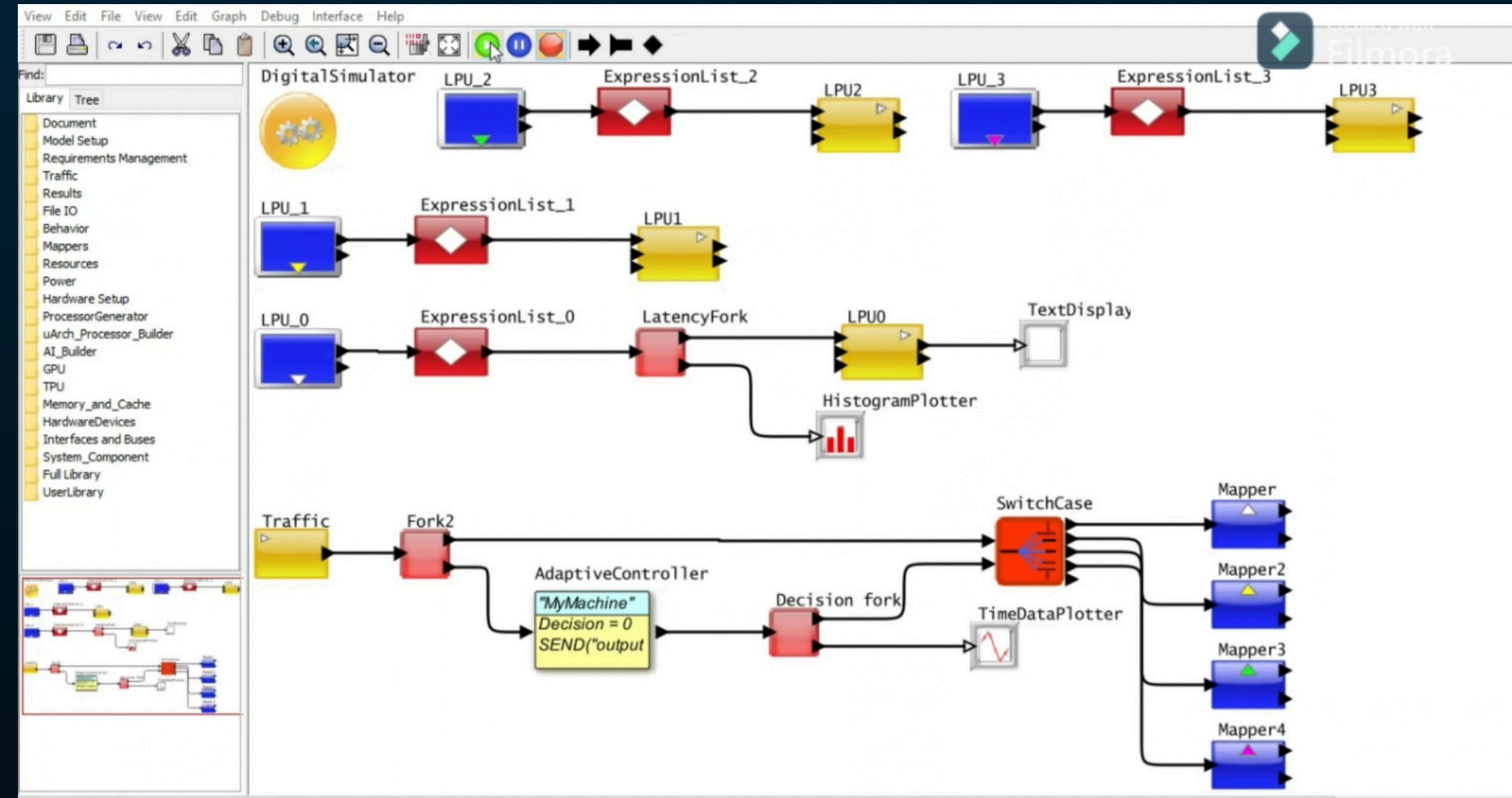


Faster processing

SYSTEM ARCHITECTURE

Adaptive Traffic Management Flow

A closed loop architecture that monitors LPU loads, makes real time decisions, and dynamically routes tasks to optimize performance



TRAFFIC INPUT

- Various Task Types
- Dynamic and bursty Workloads
- Real Life Traffic Patterns



MONITORING

- Measuring LPU Loads
- Track Congestion
- Analysis Utilization



DECISION LOGIC

- Stress aware Selection
- Choose least loaded LPU
- Real time adaption



DYNAMIC MAPPER

- Route task Adaptively
- Balance Workload
- Reduce Congestion



LPUs (Processing Units)

- Multiple LPUs works in parallel
- Execute assigned task
- Improves Resouce Utilization



STATISTICS & FEEDBACK

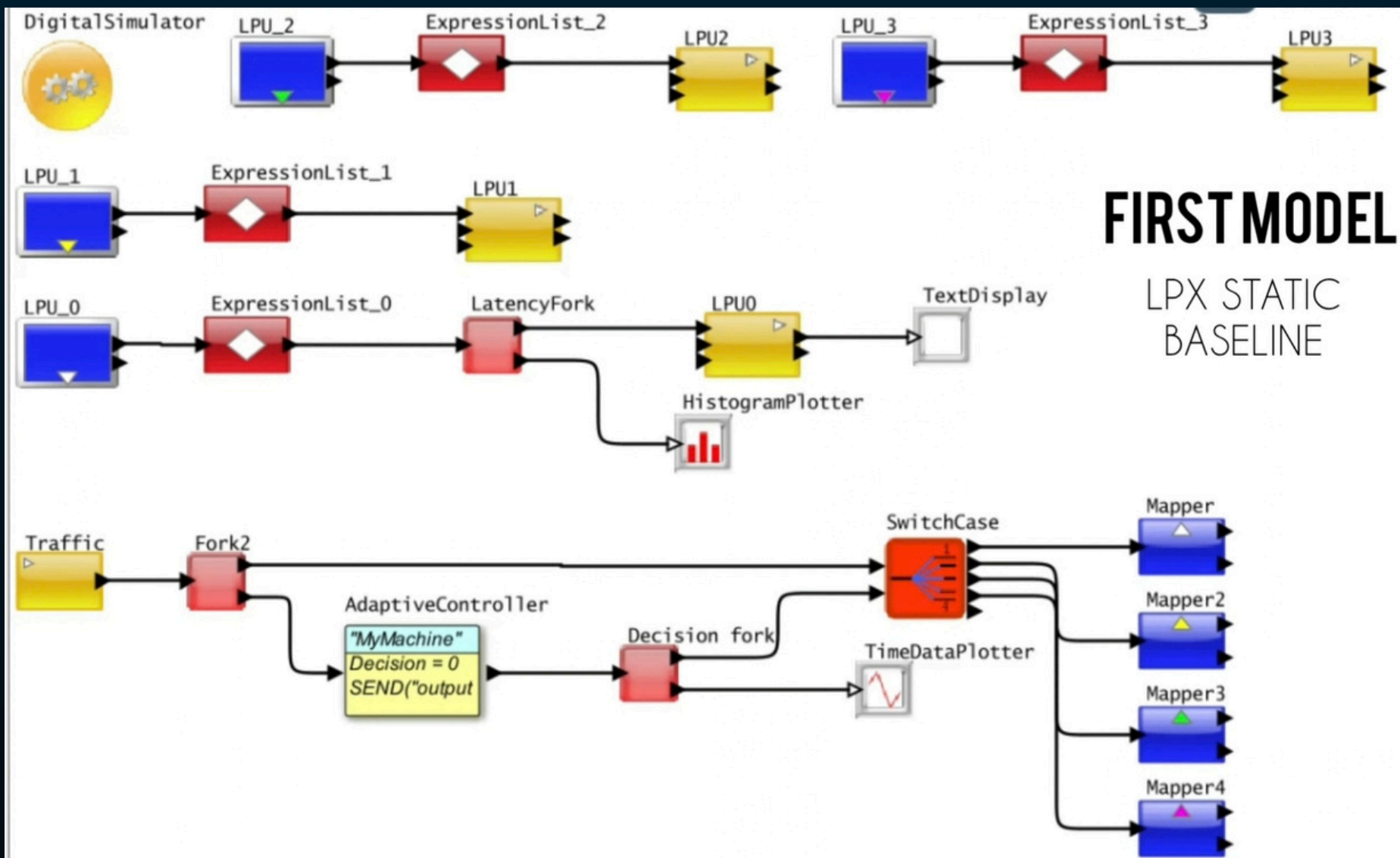
- Collects Performance Metrics
- Provides feedback for continuous optimization

Static Allocation

Tasks are mapped to LPUs using fixed routing rules without considering real-time load conditions.

Baseline for Comparison

Represents the conventional LPX architecture to evaluate the benefits of adaptive traffic management.



STATIC BASELINE MODEL

Model Overview

The LPX Static, Baseline model simulates the conventional LPX architecture where incoming tasks are mapped to LPUs using predefined static rules. The model includes a task generator, a static mapper, multiple LPUs (System_Resource blocks) and statistics modules to measure performance metrics.

Traffic Generation

- Generates incoming tasks with varying arrival patterns
- Models real-world workload behaviour

Static Mapper

- Uses fixed routing rules
- Does not consider real-time LPU load

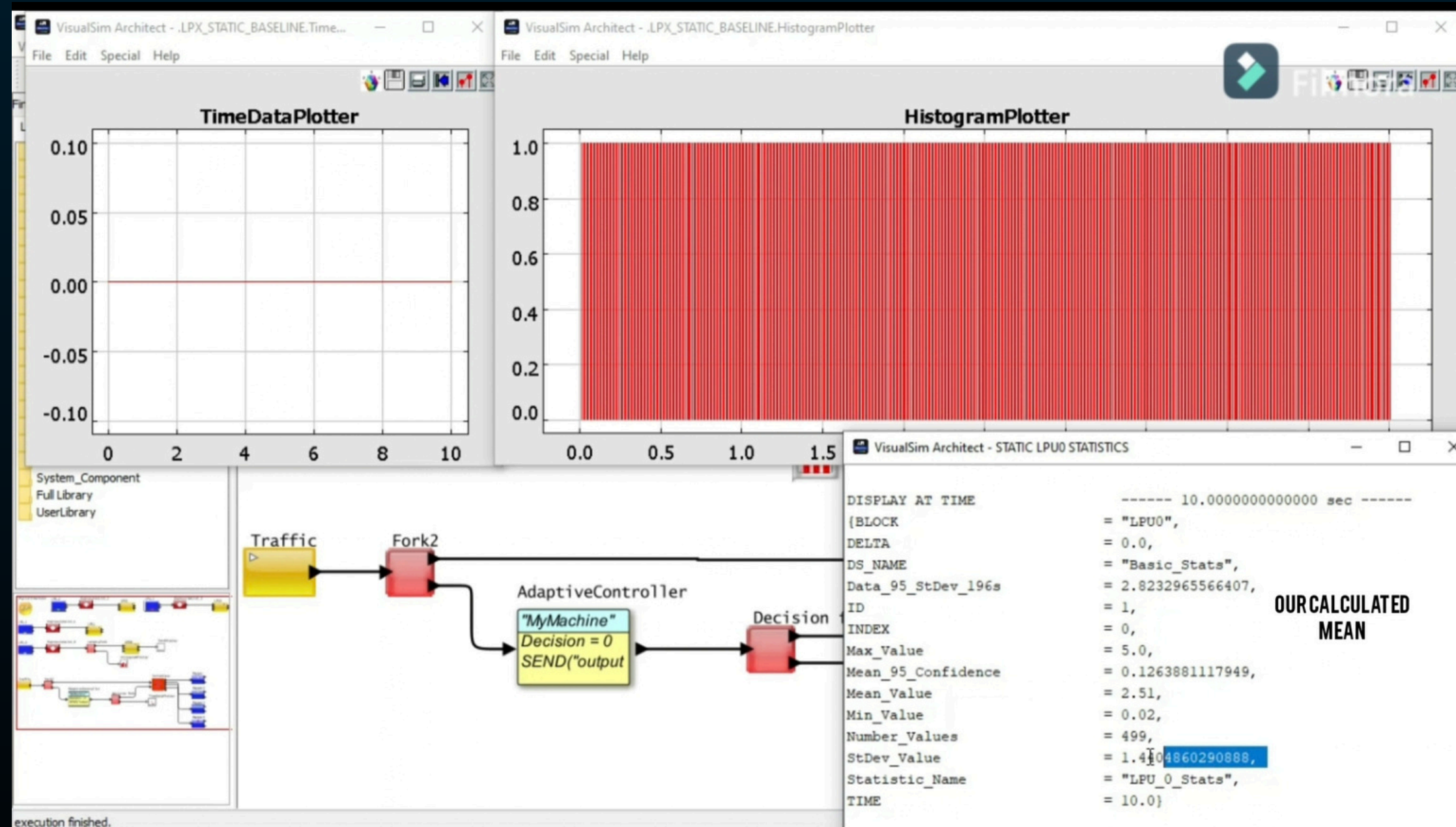
LPUs

- Multiple System_Resource blocks (LPU 0 – LPU 3)
- Execute assigned tasks in parallel

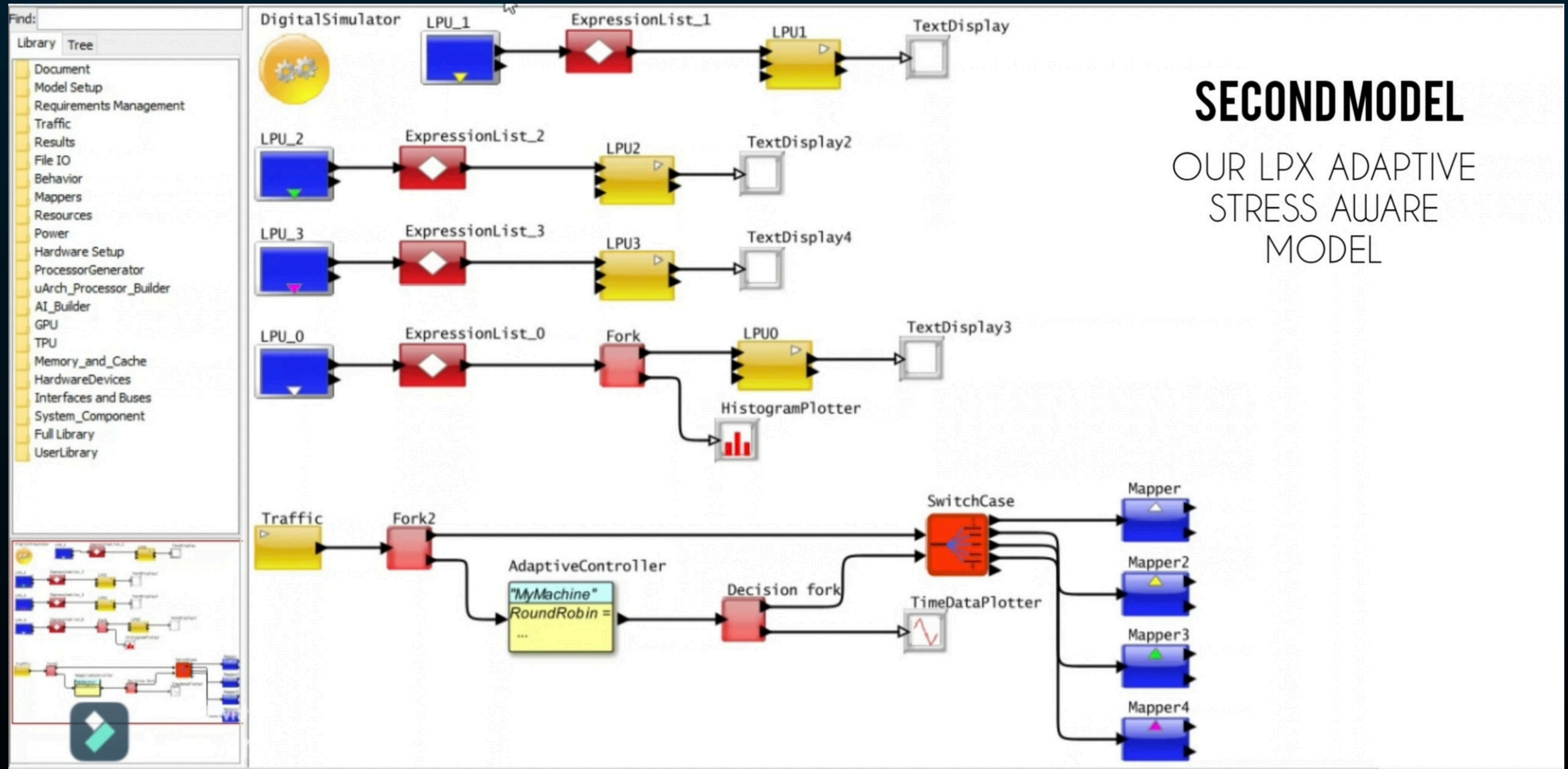
Statistics Collection

- Measures LPU utilization
- Tracks latency and throughput
- Used as a baseline to compare with the adaptive model

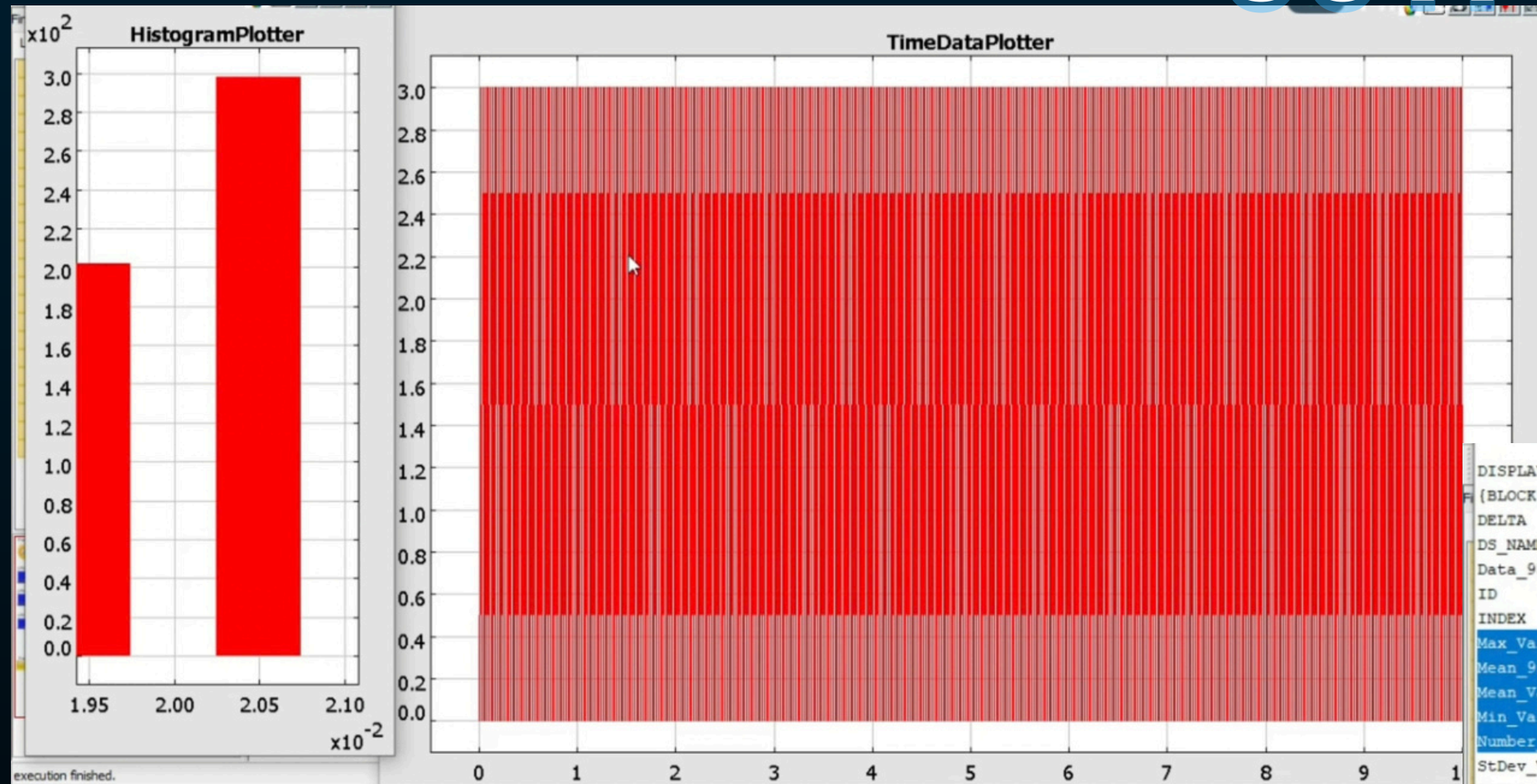
STATIC BASELINE MODEL OUTPUT



ADAPTIVE LPX STRESS AWARE MODEL



ADAPTIVE LPX STRESS AWARE MODEL OUTPUT



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DELTA                     = 0.0,
DS_NAME                   = "Basic Stats",
Data_95_StDev_196s       = 1.1469186827269E-9,
ID                         = 1,
INDEX                     = 0,
Max_Value                 = 0.02,
Mean_95_Confidence       = 5.1343131613719E-11,
Mean_Value               = 0.02,
Min_Value                 = 0.02,
Number_Values             = 499,
StDev_Value               = 5.8517421242306E-10,
Statistic_Name            = "LPU_0_Stats",
TIME                      = 10.0}
```

```
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ID                         = 1,
INDEX                     = 0,
Max_Value                 = 0.005,
Mean_95_Confidence       = 1.2797294814817E-11,
Mean_Value               = 0.005,
Min_Value                 = 0.005,
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StDev_Value               = 1.4600096599955E-10,
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```

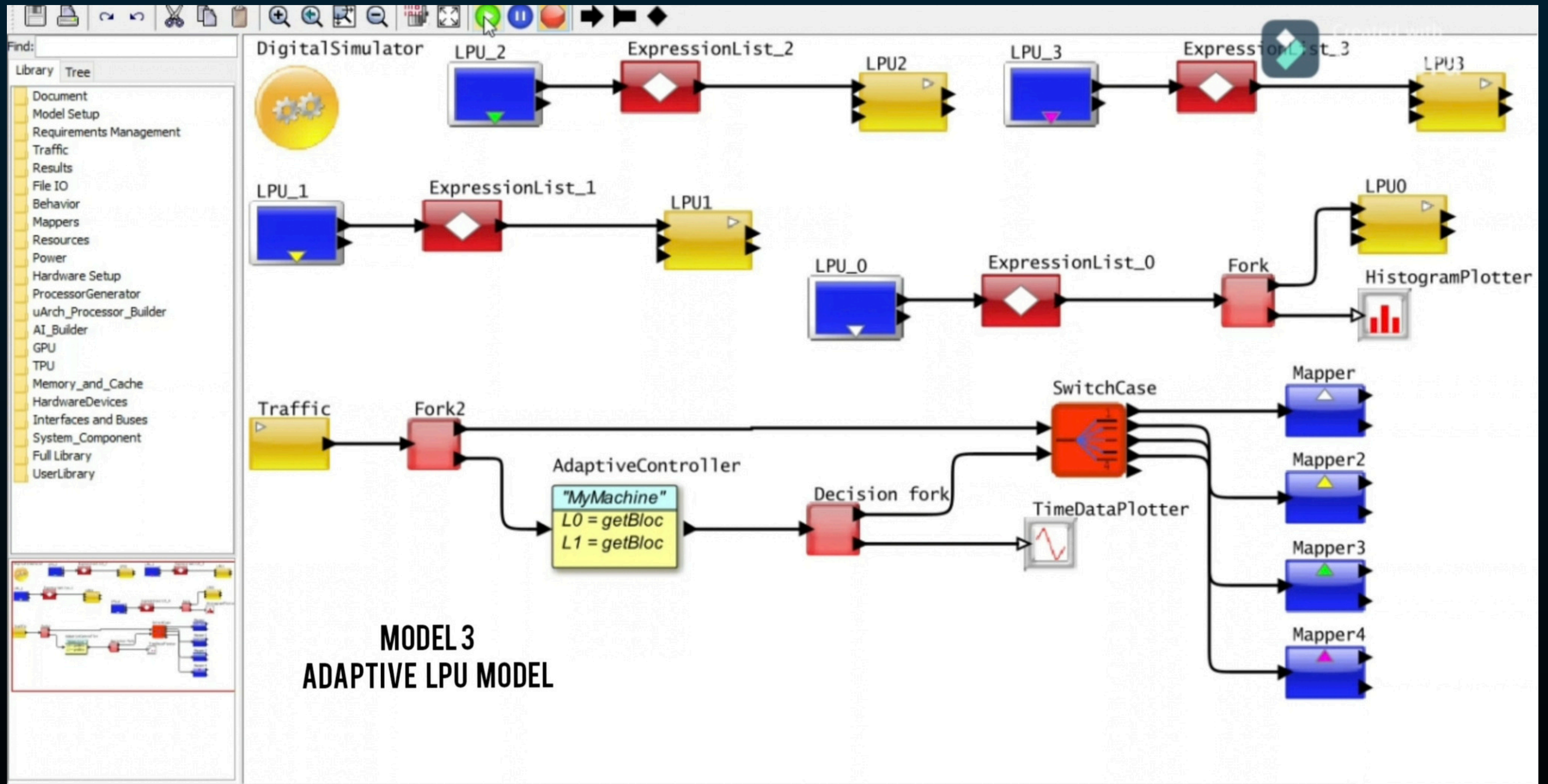
VisualSim Architect - LPU3 Statistics

```
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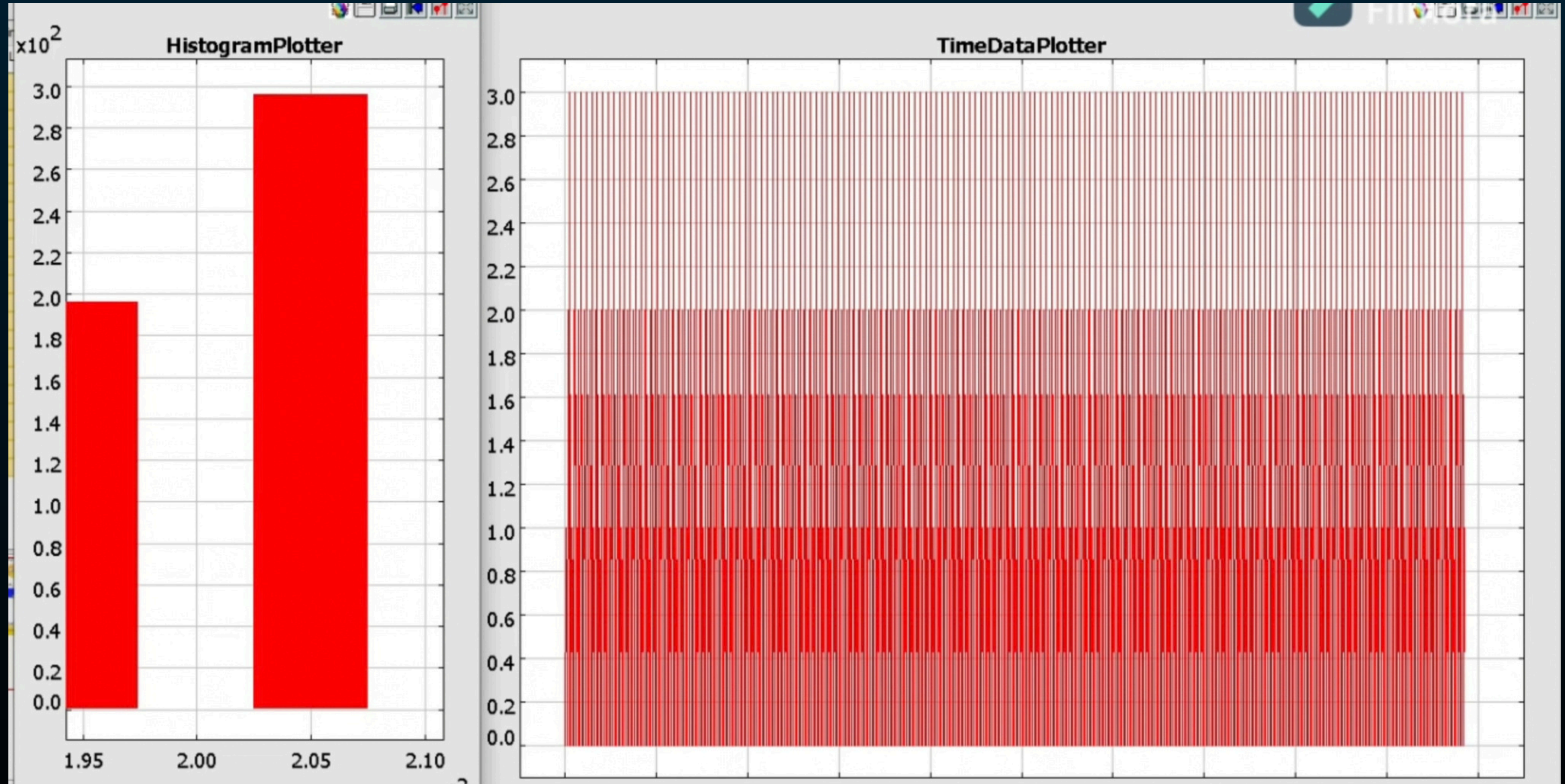
VisualSim Architect - LPU2 Statistics

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ID                         = 1,
INDEX                     = 0,
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Mean_95_Confidence       = 1.1682278408532E-11,
Mean_Value               = 0.005,
Min_Value                 = 0.005,
Number_Values             = 500,
StDev_Value               = 1.332800374925E-10,
Statistic_Name            = "LPU_2_Stats",
TIME                      = 10.0}
```

ADAPTIVE LPX MODEL



ADAPTIVE LPX MODEL OUTPUT



KEY INNOVATION

Closed-Loop Adaptation: From Fixed Routing → Intelligent, Adaptive Routing

Conventional LPX Architecture

- Uses static resource mapping
- Tasks follow predefined routing paths
- Cannot react to changing LPU workload / congestion
- May cause resource imbalance and increased latency

Our Adaptive Approach

- Continuously monitors LPU load
- Detects congestion / resource stress
- Dynamically redirects tasks to a suitable LPU
- Closed feedback loop:
Monitor → Decide → Redirect → Re-monitor

Monitor



Decide



Redirect



Re-monitor

WHY IT MATTERS

Instead of assuming the system will remain balanced, the architecture continuously adapts to its current state.

Key benefit: Better load balancing + lower congestion + improved throughput + reduced latency

Conclusion

- Developed a stress-aware adaptive LPX architecture using VisualSim.
- Replaced fixed routing with dynamic resource selection.
- Adaptive mapping responds to real-time LPU utilization.
- Demonstrates improved resource utilization, congestion handling, and system performance compared with the static baseline.
- Establishes a foundation for self-optimizing SoC architectures.

FUTURE SCOPE

Future Enhancement	Purpose
Congestion Prediction	Predict overloaded LPUs before congestion occurs
ML-Based Decisions	Learn optimal routing from workload patterns
Thermal-Aware Scheduling	Avoid thermally stressed resources
Power-Aware Mapping	Reduce energy consumption through intelligent task placement
Larger LPX Systems	Scale the approach to more LPUs and complex SoCs

CONCLUSION & FUTURE SCOPE

Future Vision

- Monitor → Predict → Decide → Adapt → Optimize

THANK YOU TO OUR FRIENDS WHO HAVE HELPED AS IN DOING OUR PROJECT

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